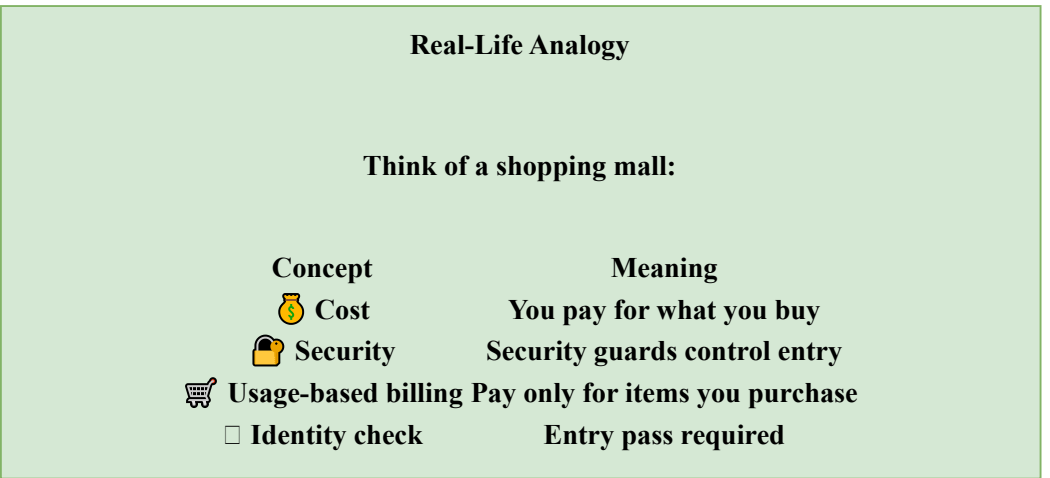
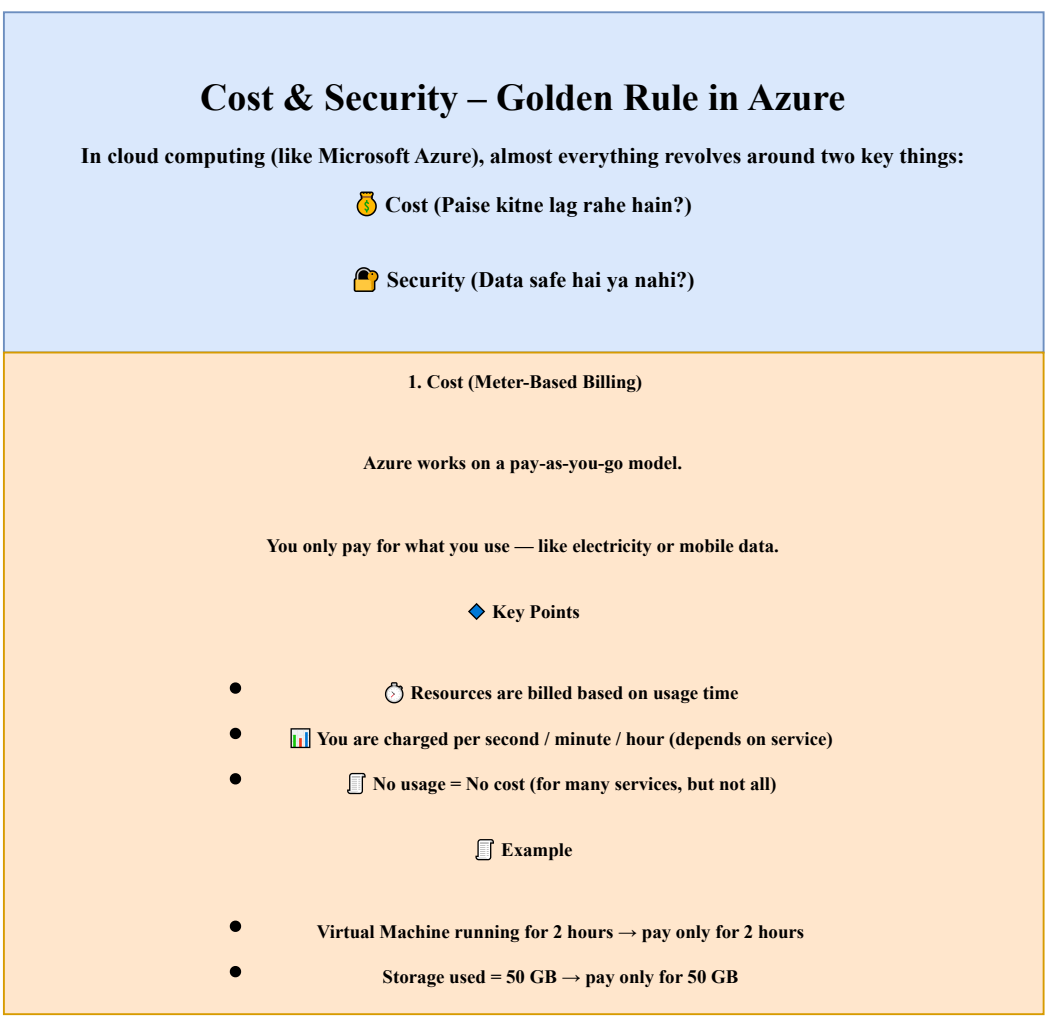
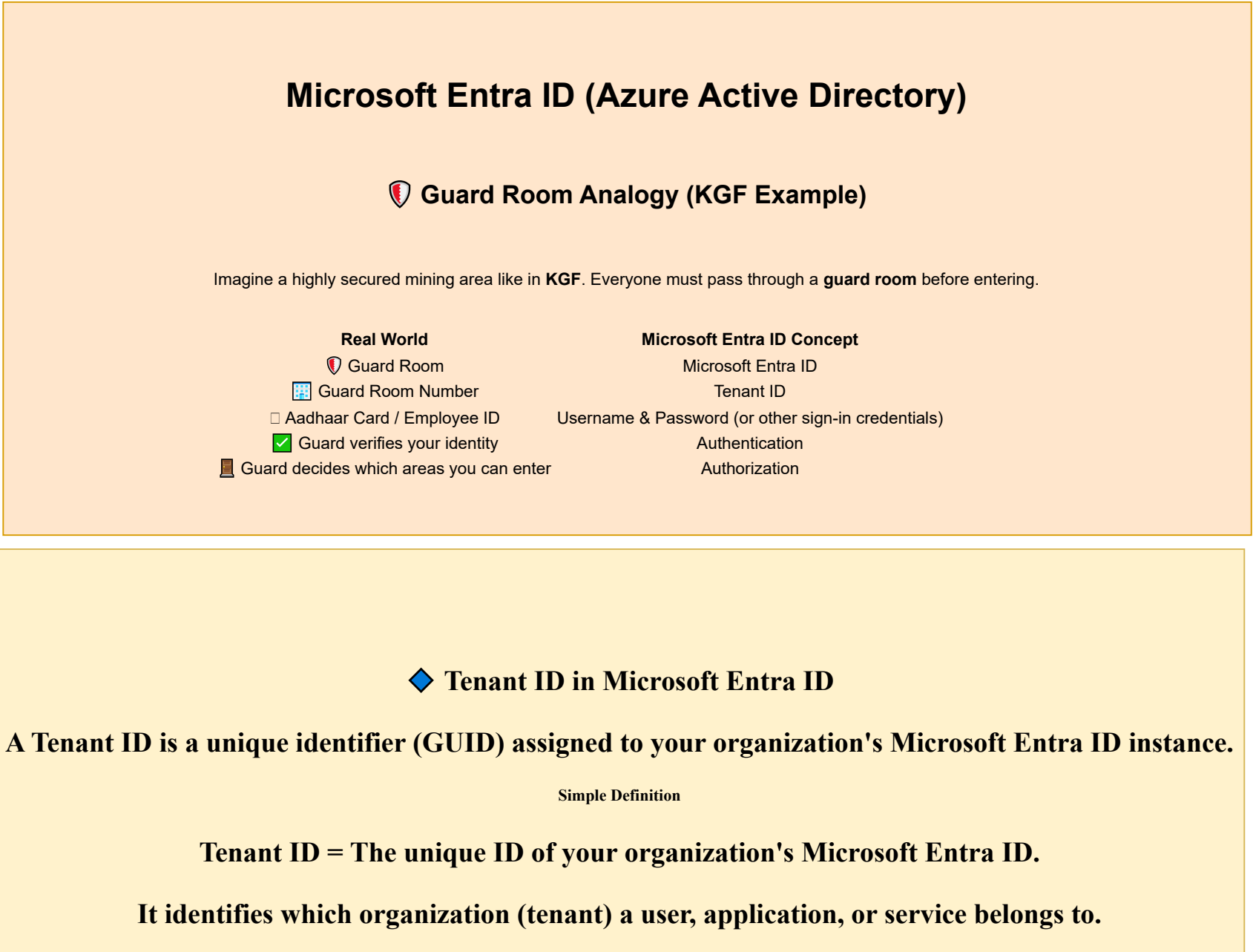
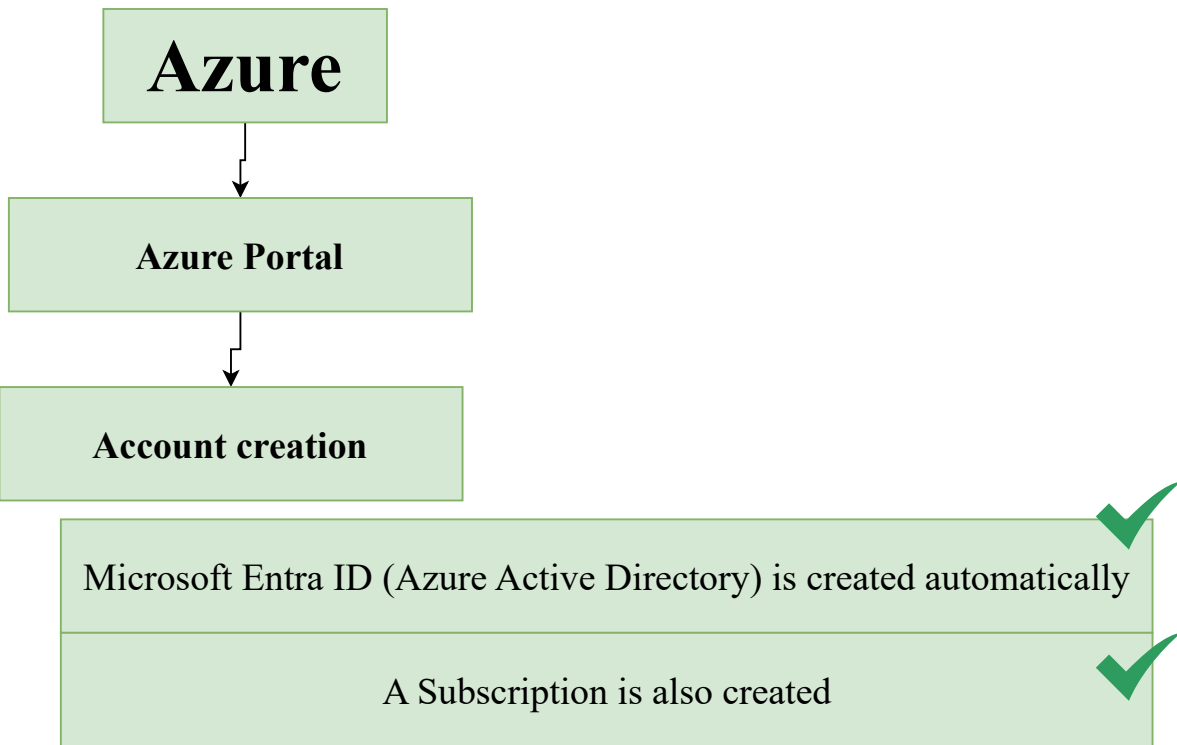
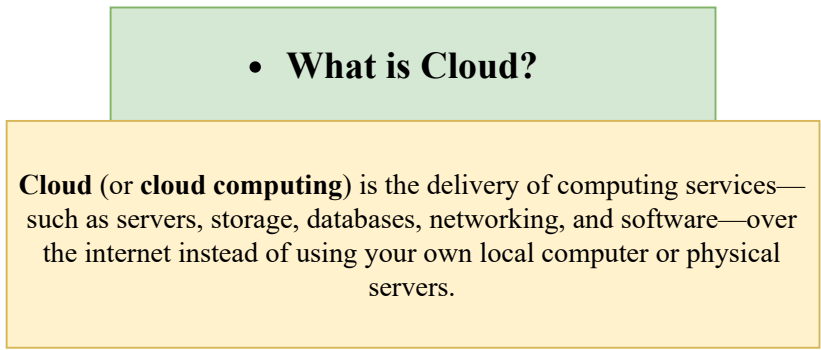
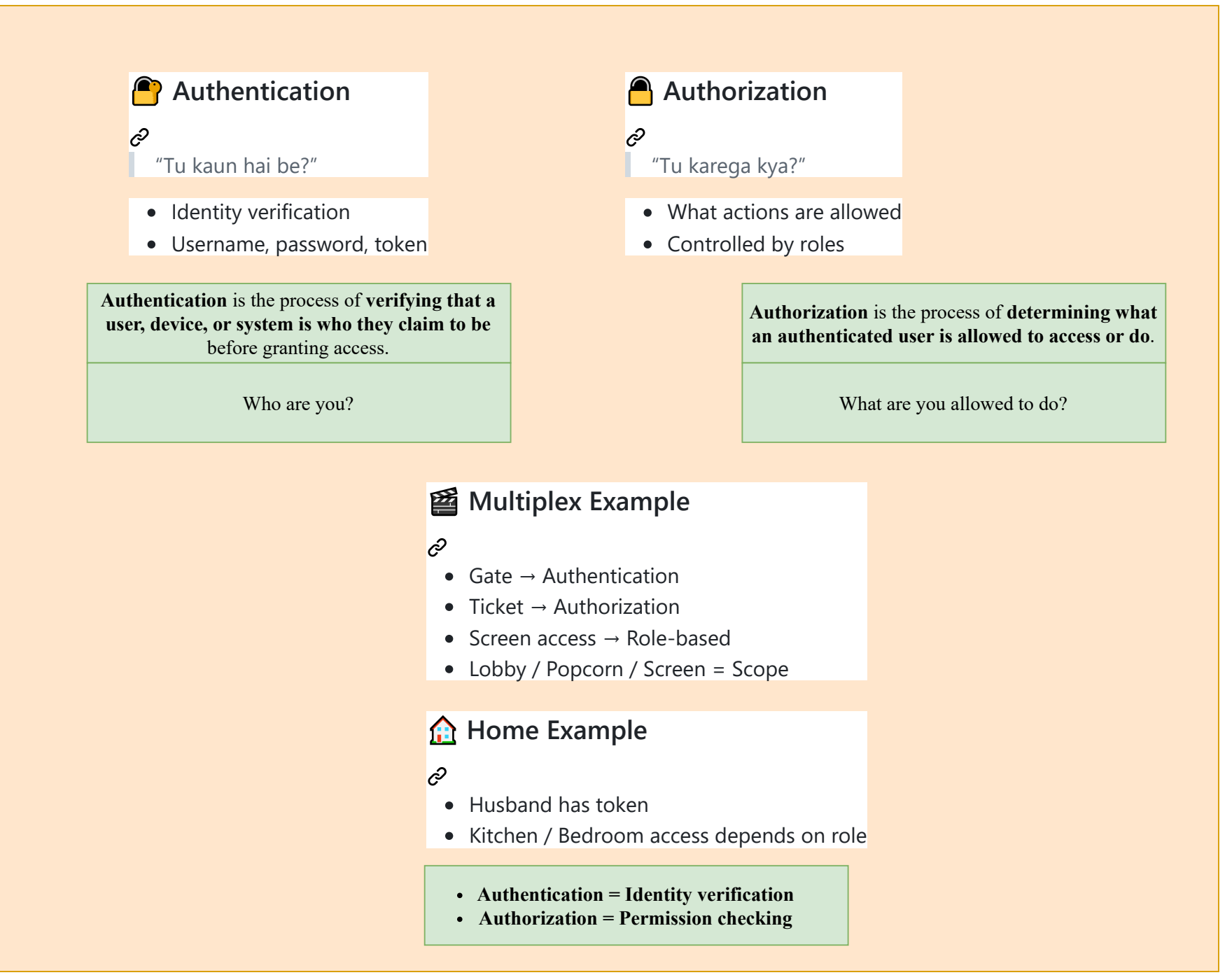
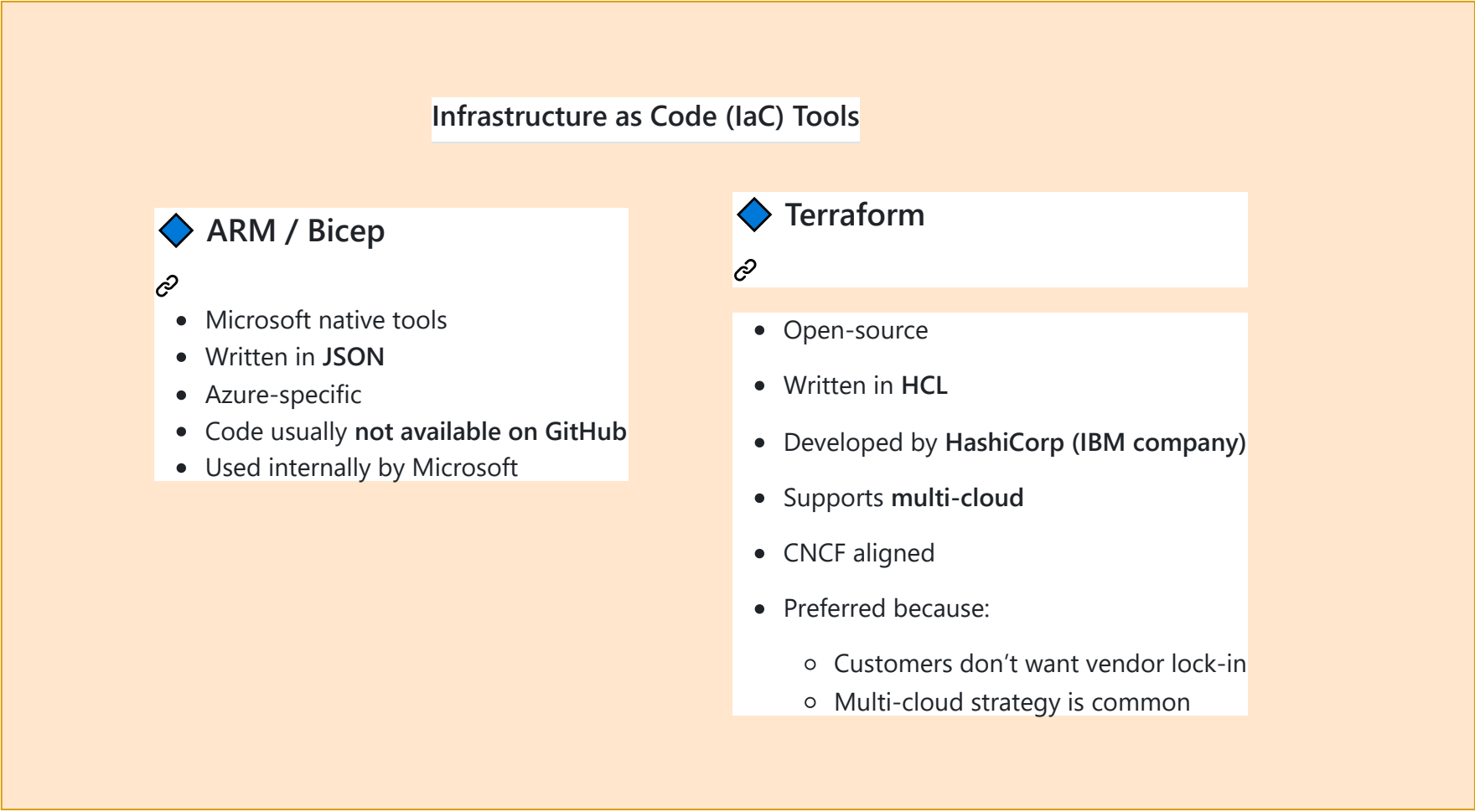
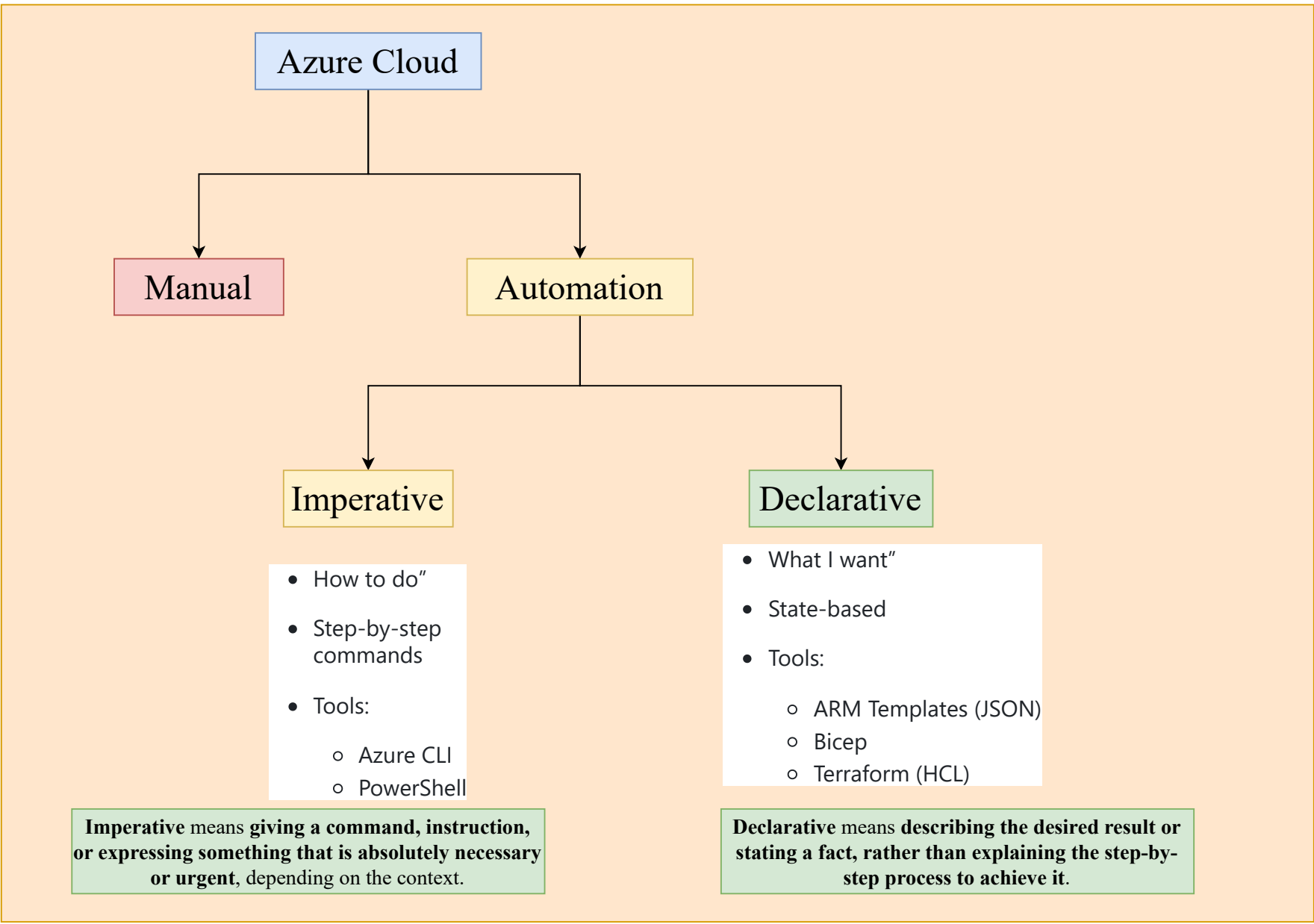
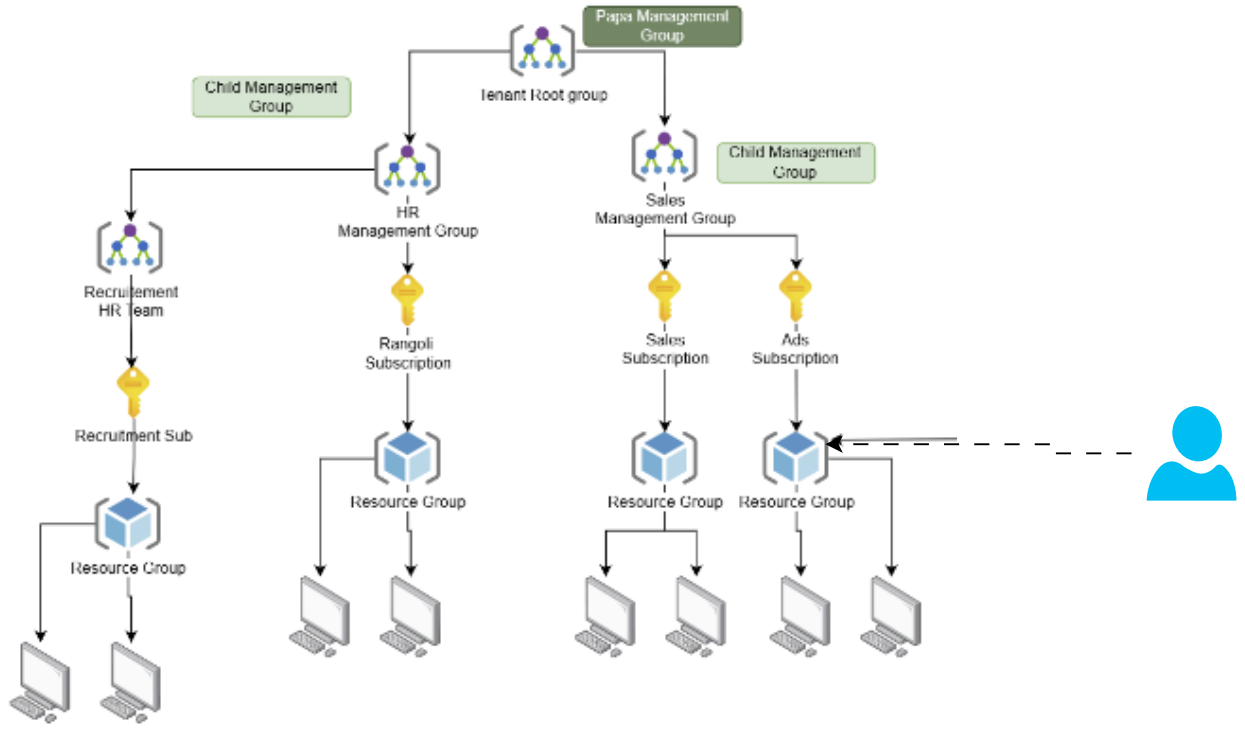
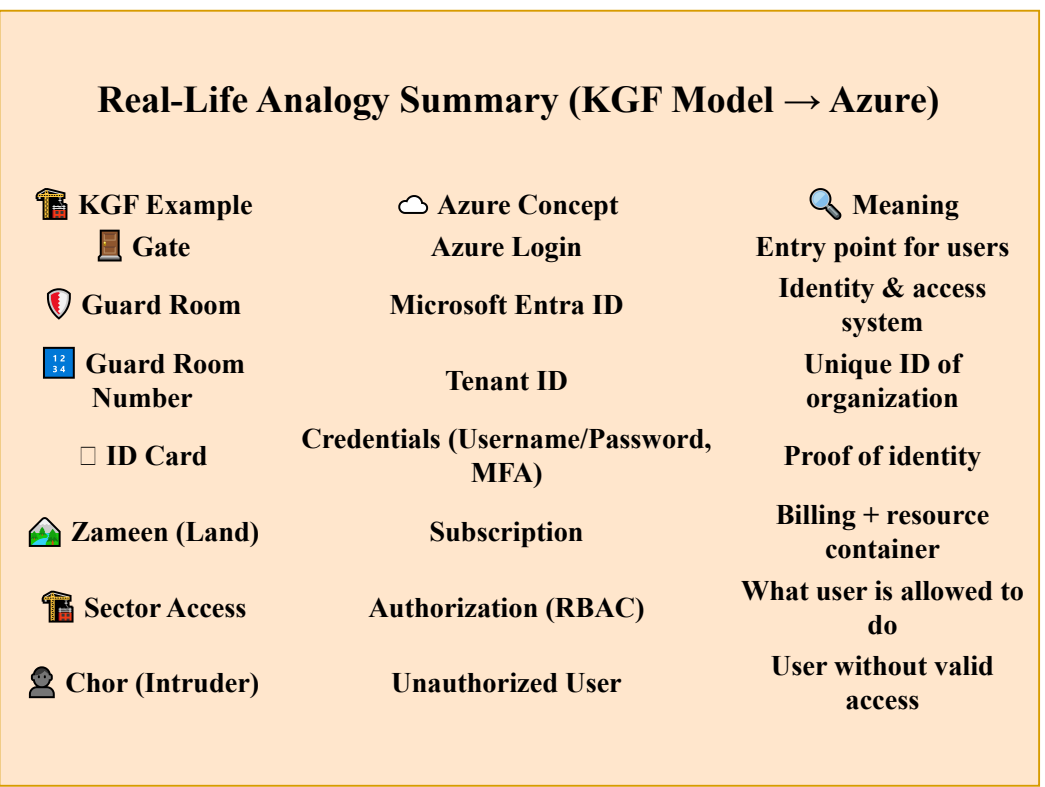




In cloud computing, cost and security are the two most important factors. Azure uses a meter-based billing system where users are charged based on resource usage. At the same time, security is enforced using identity management, authentication, authorization, and role-based access control to protect data and resources.



Example

Tenant ID:
0f7010fd-209e-4344-8457-043ffb37143b

Entra ID ki ID = Tenant ID

Users in Microsoft Entra ID

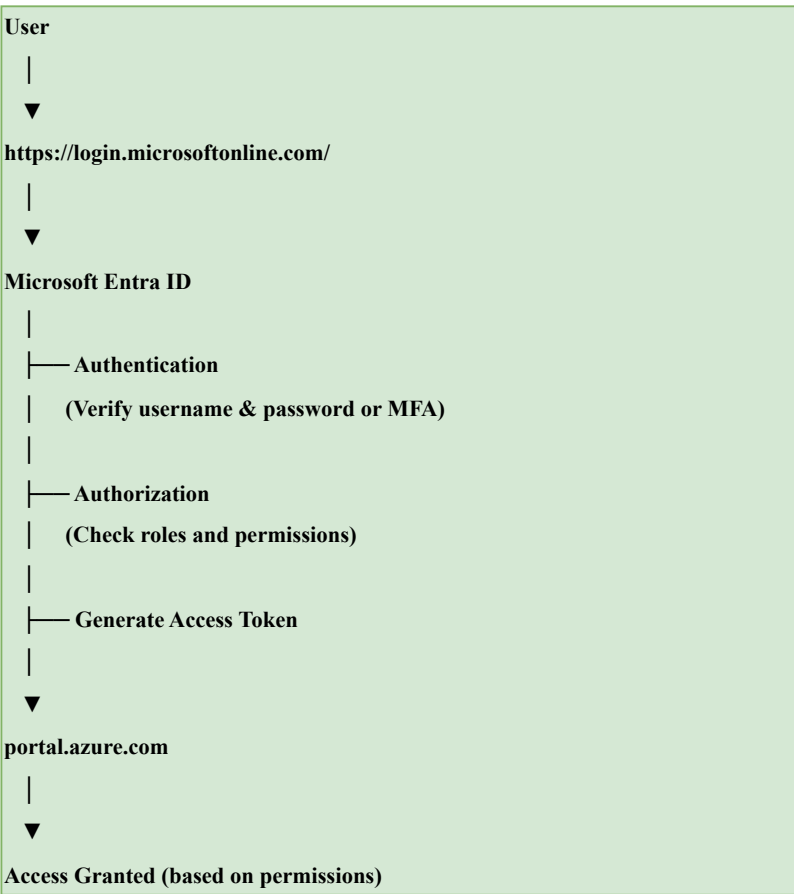
Users in Microsoft Entra ID represent the identities of people who can sign in to an organization's cloud resources.

Example Users

User	Role
Krishna Sharma	Global Administrator
Abhi Engineer	Engineer (User)
Suraj	User

Each User Has

- Name
- Email/User Principal Name (UPN)
- Password or other sign-in method
- Role (e.g., Global Administrator, User)
- Permissions to access resources
- Unique Object ID



Token-Based Access in Microsoft Entra ID

What is a Token?

A token is a temporary digital proof of authentication issued by Microsoft Entra ID after a user successfully signs in.

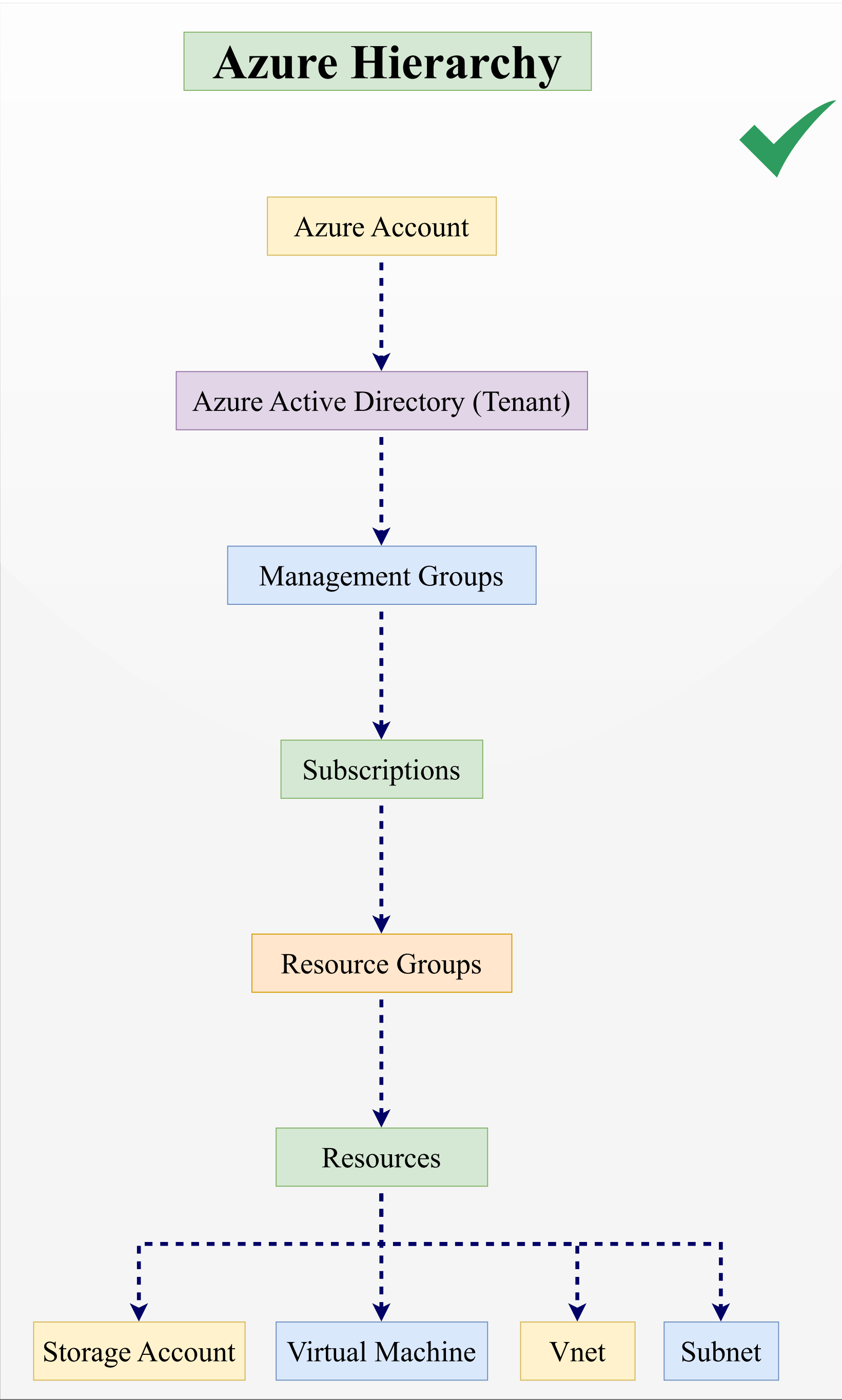
Token = A temporary digital pass that proves you have already been authenticated.

What is a Subscription?

An Azure Subscription is a logical container used to organize Azure resources and manage billing, usage, and access.

Subscription = A piece of land (Zameen) where you build your Azure resources.

All Azure resources (Virtual Machines, Storage Accounts, Databases, etc.) are created inside a subscription.



Resources & Resource Access in Azure

What are Azure Resources?

In Microsoft Azure, resources are the actual services you create and use to build applications.

Resources = The real services running in the cloud.

Examples of Azure Resources

- Virtual Machines – Run servers in the cloud
- Storage Accounts – Store files, images, backups
- Databases – Store structured data (SQL, NoSQL)
- Networking – Virtual Networks (VNet), Load Balancers, Firewalls

Azure Concept KGF Analogy	
Virtual Machine	Machine used in mining
Storage Account	Warehouse for storing gold
Database	Record book of operations
Networking	Roads and security system

How Access Works

Access to Azure resources is controlled by Microsoft Entra ID using permissions.

Access depends on:

1. Role (RBAC - Role Based Access Control)

Roles define what actions you can perform.

Examples:

- Owner → Full control
- Contributor → Can create and modify resources
- Reader → Can only view resources

Group

Users can be added to groups to manage permissions easily.

Example:

- Dev Team Group → Access to development resources
- Admin Group → Full access

Instead of giving access to each user, you assign permissions to the group.